

Lab Project:
**Network reconnaissance and
enumeration using Nmap**

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1. Lab Task Objectives / Description:

The objective of this lab is to understand and practically implement **network reconnaissance and enumeration** using **Nmap**. The lab focuses on identifying live hosts, discovering open ports, detecting running services and operating systems, performing vulnerability scans, and analyzing scan traffic using Wireshark. This lab strengthens the understanding of how attackers perform reconnaissance and how such activities can be detected.

2. Software's / Tools Required:

- Hardware: PC, Internet access
- Software: VMWare, Nmap
- VM's: Ubuntu (target-Host Only), Kali(attacker-Host only)

3. Lab Environment and Setup:

Two systems were used to perform this lab in a controlled environment. Kali Linux was used as the attacker machine, while a personal Windows PC was used as the target system. Since the Windows system is owned by the experimenter and operated within a private network, the scanning activity was ethical and aligned with academic lab guidelines. The Windows operating system provided multiple active services, making it suitable for demonstrating Nmap scanning techniques.

4. Lab Procedure & Results:

PART 1: Initial Setup

STEPS:

1. Create VM:

- Kali (Attacker): Host-only

It is important to set both the network adapters as Host only to keep the network isolated from the rest of the network.

2. Find IP addresses:

- Type command: *ip a* on Kali machine and *ipconfig* on windows CMD

to find out the IP address

```
hadiya@kali: ~  
File Actions Edit View Help  
(hadiya@kali)-[~]  
$ ip a  
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group def  
ault qlen 1000  
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00  
    inet 127.0.0.1/8 scope host lo  
        valid_lft forever preferred_lft forever  
    inet6 ::1/128 scope host noprefixroute  
        valid_lft forever preferred_lft forever  
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP g  
roup default qlen 1000  
    link/ether 00:0c:29:9d:d8:33 brd ff:ff:ff:ff:ff:ff  
    inet 192.168.10.129/24 brd 192.168.10.255 scope global dynamic noprefixro  
ute eth0  
        valid_lft 1325sec preferred_lft 1325sec  
    inet6 fe80::20c:29ff:fe9d:d833/64 scope link noprefixroute  
        valid_lft forever preferred_lft forever  
(hadiya@kali)-[~]
```

Kali Machine

```
Command Prompt  
Connection-specific DNS Suffix . :  
Ethernet adapter VMware Network Adapter VMnet1:  
Connection-specific DNS Suffix . :  
Link-local IPv6 Address . . . . . : fe80::68bf:7aa9:d652:7135%6  
IPv4 Address. . . . . : 192.168.10.1  
Subnet Mask . . . . . : 255.255.255.0  
Default Gateway . . . . . :  
Ethernet adapter VMware Network Adapter VMnet8:  
Connection-specific DNS Suffix . :  
Link-local IPv6 Address . . . . . : fe80::f30a:23e4:adf1:9f16%18  
IPv4 Address. . . . . : 192.168.205.1  
Subnet Mask . . . . . : 255.255.255.0  
Default Gateway . . . . . :  
Wireless LAN adapter Wi-Fi 3:  
Connection-specific DNS Suffix . :  
Link-local IPv6 Address . . . . . : fe80::214d:761a:ee4f:d9a6%16  
IPv4 Address. . . . . : 172.28.1.205  
Subnet Mask . . . . . : 255.255.224.0  
Default Gateway . . . . . : 172.28.1.1  
Ethernet adapter Ethernet:  
Media State . . . . . : Media disconnected
```

Windows Machine

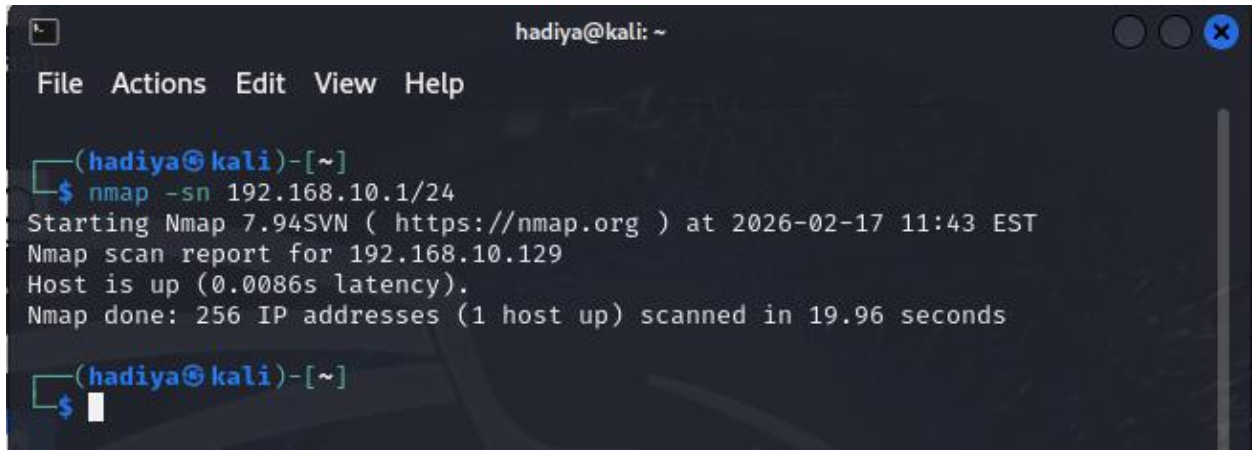
Note: The Windows firewall was temporarily disabled during the experiment to allow ICMP and TCP probes. The firewall was re-enabled after completion of the lab.

PART 2: Nmap Tasks

Task 01: Discover Live Hosts (Ping Sweep)

Run the following command to find active machines in your network:

`nmap -sn 192.168.10.0/24`

A screenshot of a terminal window titled 'hadiya@kali: ~'. The window has a menu bar with 'File', 'Actions', 'Edit', 'View', and 'Help'. The terminal shows a command prompt where the user has entered 'nmap -sn 192.168.10.1/24'. The output of the command is displayed, indicating that Nmap 7.94SVN is running at 2026-02-17 11:43 EST. It shows a scan report for 192.168.10.129, stating 'Host is up (0.0086s latency)' and 'Nmap done: 256 IP addresses (1 host up) scanned in 19.96 seconds'. The prompt returns to the user's shell.

The Windows target machine appeared as online, confirming network connectivity between the attacker and target systems.

Target Machine Configuration

Parameter	Details
Operating System	Microsoft Windows
Role	Target
Network Type	Private Network
Services Detected	SMB, RPC, NetBIOS
Firewall	Temporarily disabled for lab

Task 02: Perform a Basic Port Scan

Scan for open ports on a specific target:

`nmap -sT -Pn 192.168.10.1`

- Performs full TCP handshake
- Detects open TCP ports

```
(hadiya@kali)-[~]
$ nmap -sT 192.168.10.1
Starting Nmap 7.94SVN ( https://nmap.org ) at 2026-02-17 11:46 EST
Note: Host seems down. If it is really up, but blocking our ping probes, try
-Pn
Nmap done: 1 IP address (0 hosts up) scanned in 3.13 seconds

(hadiya@kali)-[~]
$ nmap -sT -Pn 192.168.10.1
Starting Nmap 7.94SVN ( https://nmap.org ) at 2026-02-17 11:47 EST
Nmap scan report for 192.168.10.1
Host is up (0.0026s latency).
Not shown: 999 filtered tcp ports (no-response)
PORT      STATE SERVICE
3306/tcp  open  mysql

Nmap done: 1 IP address (1 host up) scanned in 19.78 seconds
```

Finding:

During scanning, the Windows target blocked ICMP echo requests, causing Nmap to initially report the host as down. The **-Pn** option was used to bypass host discovery and force the scan, successfully revealing open TCP ports. This reflects real world scenarios where firewalls block ping but allow service traffic.

Task 3: Perform a Stealth Scan (SYN Scan)

Use a stealth scan to avoid detection:

`Sudo nmap -sS -Pn 192.168.10.1`

```

(hadiya@kali)-[~]
$ sudo nmap -sS -Pn 192.168.10.1
[sudo] password for hadiya:
Starting Nmap 7.94SVN ( https://nmap.org ) at 2026-02-17 11:51 EST
Nmap scan report for 192.168.10.1
Host is up (0.00083s latency).
Not shown: 999 filtered tcp ports (no-response)
PORT      STATE SERVICE
3306/tcp  open  mysql
MAC Address: 00:50:56:C0:00:01 (VMware)

Nmap done: 1 IP address (1 host up) scanned in 18.29 seconds

```

Open ports were detected without establishing full connections, demonstrating how SYN scans are less likely to be logged by intrusion detection systems.

Task 4: Detect Operating System and Service Versions

Identify the OS and running services:

Sudo nmap -O -sV -Pn 192.168.10.1

- Detects operating system
- Detects service versions

```

(hadiya@kali)-[~]
$ sudo nmap -O -sV -Pn 192.168.10.1
Starting Nmap 7.94SVN ( https://nmap.org ) at 2026-02-17 11:52 EST
Nmap scan report for 192.168.10.1
Host is up (0.0012s latency).
Not shown: 999 filtered tcp ports (no-response)
PORT      STATE SERVICE VERSION
3306/tcp  open  mysql    MySQL (unauthorized)
MAC Address: 00:50:56:C0:00:01 (VMware)
Warning: OSScan results may be unreliable because we could not find at least
1 open and 1 closed port
Device type: general purpose
Running (JUST GUESSING): Microsoft Windows 11|10 (92%), FreeBSD 6.X (88%)
OS CPE: cpe:/o:freebsd:freebsd:6.2 cpe:/o:microsoft:windows_10
Aggressive OS guesses: Microsoft Windows 11 21H2 (92%), FreeBSD 6.2-RELEASE (
88%), Microsoft Windows 10 (87%)
No exact OS matches for host (test conditions non-ideal).
Network Distance: 1 hop

OS and Service detection performed. Please report any incorrect results at ht
tps://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 25.00 seconds

```

Nmap successfully identified the target operating system as Microsoft Windows. Service version detection revealed Microsoft RPC and SMB services, confirming the effectiveness of OS fingerprinting and service enumeration techniques.

Task 5: Perform an Aggressive Scan

Gather maximum information about the target:

Sudo nmap -A 192.168.10.1

```
(hadiya@kali)-[~]
$ sudo nmap -A -Pn 192.168.10.1
Starting Nmap 7.94SVN ( https://nmap.org ) at 2026-02-17 11:56 EST
Nmap scan report for 192.168.10.1
Host is up (0.00098s latency).
Not shown: 999 filtered tcp ports (no-response)
PORT      STATE SERVICE VERSION
3306/tcp  open  mysql   MySQL (unauthorized)
MAC Address: 00:50:56:C0:00:01 (VMware)
Warning: OSScan results may be unreliable because we could not find at least
1 open and 1 closed port
Device type: general purpose
Running (JUST GUESSING): Microsoft Windows 11|10 (92%), FreeBSD 6.X (88%)
OS CPE: cpe:/o:freebsd:freebsd:6.2 cpe:/o:microsoft:windows_10
Aggressive OS guesses: Microsoft Windows 11 21H2 (92%), FreeBSD 6.2-RELEASE (
88%), Microsoft Windows 10 (87%)
No exact OS matches for host (test conditions non-ideal).
Network Distance: 1 hop

TRACEROUTE
HOP RTT      ADDRESS
1   0.98 ms  192.168.10.1

OS and Service detection performed. Please report any incorrect results at ht
tps://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 29.68 seconds
```

The aggressive scan combined:

- OS detection
- Service enumeration
- Script scanning
- Traceroute

This provided comprehensive information about the target system in a single scan.

Task 6: Scan for Vulnerabilities

Use Nmap Scripting Engine (NSE) to detect vulnerabilities:

nmap --script vuln 192.168.10.1

```
(hadiya@kali)-[~]  
$ sudo nmap --script vuln -Pn 192.168.10.1  
Starting Nmap 7.94SVN ( https://nmap.org ) at 2026-02-17 12:00 EST  
Nmap scan report for 192.168.10.1  
Host is up (0.0014s latency).  
Not shown: 999 filtered tcp ports (no-response)  
PORT      STATE SERVICE  
3306/tcp  open  mysql  
MAC Address: 00:50:56:C0:00:01 (VMware)  
  
Nmap done: 1 IP address (1 host up) scanned in 19.98 seconds
```

The vulnerability scan executed Nmap NSE scripts against detected Windows services. While no critical vulnerabilities were identified, several informational scripts related to SMB configuration and RPC services were displayed. This demonstrated how vulnerability scanners assess service exposure even in secure systems.

Task 7: Capture and Analyze Scan Traffic with Wireshark

1. Open Wireshark on the target machine.
2. Start capturing packets on the network interface.
3. Run an Nmap scan from the attacker machine.
4. Stop the capture and analyze TCP SYN packets to see Nmap activity.

tcp.flags.syn == 1

Capturing from Wi-Fi 3

File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help

tcp.flags.syn == 1

No.	Time	Source	Destination	Protocol	Length	Info
9	0.020512	52.123.245.131	172.28.1.205	TCP	66	443 → 51262 [SYN, ACK] Seq=0 Ack=1 Win=65535 Len=0 MSS=1380 WS=256 SACK_PERM
15	0.026011	172.28.1.205	20.190.146.38	TCP	66	65453 → 443 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 WS=256 SACK_PERM
16	0.106025	172.28.1.205	104.26.10.240	TCP	66	61478 → 443 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 WS=256 SACK_PERM
23	0.224255	20.190.146.38	172.28.1.205	TCP	66	443 → 65453 [SYN, ACK] Seq=0 Ack=1 Win=65535 Len=0 MSS=1380 WS=256 SACK_PERM
28	0.231150	172.28.1.205	40.119.213.159	TCP	66	56162 → 443 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 WS=256 SACK_PERM
31	0.408919	104.26.10.240	172.28.1.205	TCP	66	443 → 61478 [SYN, ACK] Seq=0 Ack=1 Win=65535 Len=0 MSS=1380 SACK_PERM WS=8192
49	0.481709	172.28.1.205	40.119.213.159	TCP	66	54941 → 443 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 WS=256 SACK_PERM
55	0.622882	40.119.213.159	172.28.1.205	TCP	66	443 → 54941 [SYN, ACK] Seq=0 Ack=1 Win=64240 Len=0 MSS=1380 SACK_PERM WS=1024
89	0.841217	172.28.1.205	104.26.10.240	TCP	66	65452 → 443 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 WS=256 SACK_PERM
122	1.046503	104.26.10.240	172.28.1.205	TCP	66	443 → 65452 [SYN, ACK] Seq=0 Ack=1 Win=65535 Len=0 MSS=1380 SACK_PERM WS=8192
144	1.121222	172.28.1.205	40.119.213.159	TCP	66	(TCP Retransmission) 54392 → 443 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 WS=256 SACK_PERM
153	1.425037	172.28.1.205	52.123.245.131	TCP	66	54392 → 443 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 WS=256 SACK_PERM
170	1.687157	172.28.1.205	52.123.245.131	TCP	66	58937 → 443 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 WS=256 SACK_PERM
217	2.426692	172.28.1.205	52.123.245.131	TCP	66	(TCP Retransmission) 58937 → 443 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 WS=256 SACK_PERM
240	2.939955	52.123.245.131	172.28.1.205	TCP	66	443 → 58937 [SYN, ACK] Seq=0 Ack=1 Win=65535 Len=0 MSS=1380 WS=256 SACK_PERM
252	2.943325	172.28.1.205	40.99.27.2	TCP	66	58938 → 443 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM
253	2.943358	172.28.1.205	40.99.27.2	TCP	66	58939 → 443 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM
263	3.011793	172.28.1.205	52.123.245.131	TCP	66	54766 → 443 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 WS=256 SACK_PERM
277	3.163309	40.99.27.2	172.28.1.205	TCP	66	443 → 58939 [SYN, ACK] Seq=0 Ack=1 Win=65535 Len=0 MSS=1380 WS=256 SACK_PERM
278	3.163309	40.99.27.2	172.28.1.205	TCP	66	443 → 58938 [SYN, ACK] Seq=0 Ack=1 Win=65535 Len=0 MSS=1380 WS=256 SACK_PERM
286	3.168608	52.123.245.131	172.28.1.205	TCP	66	443 → 54766 [SYN, ACK] Seq=0 Ack=1 Win=65535 Len=0 MSS=1380 WS=256 SACK_PERM
383	4.641220	172.28.1.205	8.8.4.4	TCP	66	59503 → 443 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 WS=256 SACK_PERM
385	4.656851	172.28.1.205	8.8.4.4	TCP	66	64627 → 443 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 WS=256 SACK_PERM
386	4.657154	172.28.1.205	8.8.8.8	TCP	66	64142 → 443 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 WS=256 SACK_PERM
388	4.699560	8.8.8.8	172.28.1.205	TCP	66	443 → 64142 [SYN, ACK] Seq=0 Ack=1 Win=65535 Len=0 MSS=1380 SACK_PERM WS=256

Frame 9: 66 bytes on wire (528 bits), 66 bytes captured (528 bits) on interface \Device\NPF_{E53C6AA2-F7...}

Ethernet II, Src: HuaweiTechno_38-43-17 (48:15:13:38:43:17), Dst: WistronNewb_cc6d:dd (8c:53:e6:cc:6d:dd)

Internet Protocol Version 4, Src: 52.123.245.131, Dst: 172.28.1.205

Transmission Control Protocol, Src Port: 443, Dst Port: 51262, Seq: 0, Ack: 1, Len: 0

0000 8c 53 e6 cc 6d dd 44 a1 91 33 43 17 00 00 45 00 S m D 3C E

0010 00 34 e5 ee 40 00 6a 06 53 2d 34 7b f5 03 ae 1c 4 0 j S-4i

0020 01 cd 01 bb c8 3e b3 8a 19 51 99 9c sd 39 00 12 > Q j9

0030 ff ff 09 bd 00 00 02 04 05 64 01 03 03 00 01 01 d

0040 04 02

Wireshark analysis showed a high number of TCP SYN packets originating from the Kali machine toward the Windows target. During SYN scans, incomplete TCP handshakes were observed, while TCP connect scans resulted in full three-way handshakes. This traffic pattern is characteristic of network reconnaissance activity.

Conclusion

In this lab, network scanning and enumeration techniques were successfully performed using Nmap in a controlled virtual environment. Various scan types were used to identify live hosts, open ports, running services, operating systems, and potential vulnerabilities. Wireshark analysis provided visibility into how Nmap interacts with the target at the packet level. This lab enhanced practical understanding of reconnaissance techniques used in real-world cyber attacks and emphasized the importance of monitoring and securing network services.